



Transitioning from hydrothermal carbonization to humification for producing artificial humic substances

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HIGHLIGHTS

- Chemical shift from acid to base enables selective biomass humification.
- Glucose and cellulose form humic products via *retro*-aldol and condensation.
- Lignin undergoes intense defragmentation under strong base.
- Solid-state NMR reveals phenolic and polyaromatic features in humic-like products.
- Artificial humic substances are fluorescent and suited for soil applications.

GRAPHICAL ABSTRACT



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ABSTRACT

Hydrothermal humification (HTH) is an emerging green route for converting biomass into artificial humic substances (A-HS), oxygen-rich macromolecules resembling natural humics and relevant for carbon sequestration and soil improvement. HTH represents the alkaline version of hydrothermal carbonization (HTC), acidic and yielding to insoluble hydrochars. Here, we investigate the pathways behind this transition by treating glucose, cellulose, and lignin at 220 °C for 4 h under controlled concentrations of alkali (KOH). Under HTH, carbohydrates undergo *retro*-aldol cleavage, yielding low solid yields and lactic acid and aldehydes that condense into phenolic/aromatic A-HS, as shown by solid-state NMR. In contrast, HTC mainly forms furanic hydrochars. Lignin under HTH undergoes near-complete depolymerization/defragmentation (<1.1 % residue), yielding oxygen-rich aromatic A-HS, whereas it remains unreacted under HTC. Overall, the resulting A-HS are aromatic, rich in oxygenated functionalities, and fluorescent, resembling natural HS and highlighting the potential of HTH for producing soil-relevant humics from biomass.

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1. Introduction

Soil is a crucial yet often overlooked strategy for addressing climate change. With ~2345 Gt of organic carbon (first three meters), soil is one of the largest global carbon pools, deeply affecting GHG emissions (Lehmann et al., 2020). In particular, soil organic carbon is a well-known indicator of soil health and resilience, with humic substances (HS) playing a central role in this regard (Rumpel et al., 2018). HS occupy up to 60–80 % of soil organic matter, regulating functions such as structure, nutrient availability, and metal complexation, while their persistence makes them carbon sinks (Lee et al., 2019). HS form through the degradation of biomass and microbial residues mediated by mineral interactions over centuries in soil (De Melo et al., 2016). They are heterogeneous supramolecular assemblies – rich in phenolic and carboxylic groups and stabilized by hydrogen bonding and hydrophobic interactions and their unique properties have driven a growing market in agriculture (as soil amendments) and in environmental remediation (Yang et al., 2021). However, commercial HS are mainly obtained from non-renewable carbon stocks (lignite, peat, weathered coal), while natural HS are impractical to recover from soils due to low yields and high impurities (Wei et al., 2022). In this context, researchers have been investigating the synthesis of HS artificial counterpart, called “artificial-HS” (A-HS). Current synthetic approaches involve the conversion of biomass via biotic or abiotic processes (Wei et al., 2022). Biotic processes – such as composting, vermicomposting, and anaerobic digestion – are inexpensive but limited by long treatment times (days to months) and low efficiency (Shao et al., 2022). Abiotic processes, primarily catalytic or hydrothermal, are more technologically demanding yet operate within hours, allowing for precise reaction control. Catalytic processes are based on the polymerization of precursors (like polyphenols and amino acids) using oxides of Mn, Si, Al, Fe, or minerals, but suffer from high costs (Nishimoto et al., 2013). Hydrothermal processes exploit the unique properties of hot pressurized water, and among them, hydrothermal humification (HTH) has recently emerged for the direct synthesis of A-HS (Ischia et al., 2024). HTH adopts an alkali controlled environment at 180–250 °C 10–45 bar to depolymerize biomass into intermediates that recondense into polymeric structures similar to natural HS (Yang et al., 2021). Recent studies have shown the effectiveness of A-HS from HTH in applications, for example, the recovery of phosphorus and heavy-metals from sludge (Zhang et al., 2020), support of soil microbial communities (Hoefle et al., 2024), and as end-products in biorefineries (Queffelec et al., 2025).

HTH still requires scientific efforts for its advancement. While most studies focused on A-HS characterization and applications, the conversion mechanisms remain little explored (Shao et al., 2023) complicit the complexity of the reaction environment and biomass heterogeneity. However, insights can be drawn from hydrothermal carbonization (HTC), the HTH acidic version where no alkali agent is used, extensively investigated in literature for producing carbon-rich hydrochar (Benavente et al., 2022). HTC shows several limitations for soil applications: hydrochar is often phytotoxic (mainly due to furans) and requires post-treatments such as aging or composting and the HTC liquid phase, the main product by mass, is usually discarded (Ischia et al., 2024). HTH builds on HTC principles, but alkali shifts the pathway from hydrochar to humic-like structures. Understanding the chemical mechanisms underlying the transition from HTC-acidic to HTH-alkali is of both scientific and technological relevance. Fundamentally, it provides a scientific base for tailoring A-HS properties, as well as a rational framework for alkali dosage based on the substrate. Then, it can be a between the HTC and humification communities so to facilitate the adoption of HTH in more technologically mature HTC systems. In this regard, a recent study examined the conversion of digestate under HTC and HTH regimes, emphasizing the role of alkali in promoting digestate humification (Marzban et al., 2024). Most of the current literature, however, has focused primarily on A-HS applications or on maximizing their yields, with mechanistic investigations being less studied and

mostly limited to HTH (Shao et al., 2023). To the best of the author's knowledge, the transition HTC-HTH has not yet been systematically investigated in terms of mechanisms. In this work, we address this gap by studying the transition from HTC to HTH through the hydrothermal treatment of model biomass compounds – glucose, cellulose, and lignin – under increasing alkali (KOH) concentrations. These substrates were chosen as representative model compounds of lignocellulosic biomass with the aim of understanding fundamental mechanisms that can serve as a starting point for more complex and heterogeneous feedstocks. All solid and liquid products – either from HTH (A-HS, comprising humic acid, fulvic acids, and humin) or from HTC (hydrochar and liquid phase) – were thoroughly characterized. Results allowed us to identify conversion pathways and key differences between products, underscoring the central role of alkalinity in enabling A-HS formation.

2. Materials and methods

2.1. Hydrothermal runs

Glucose (Sigma-Aldrich, ≥99.5 %), cellulose (microcrystalline Avicel®), and lignin (Organosolv lignin, Fraunhofer CBP) were used in powder form and hydrothermally treated in 50 mL Parr pressure vessels (Moline, IL, USA). Each vessel was loaded with 2.1 g biomass and 7 mL distilled water (biomass-to-water ratio = 0.3), ensuring glucose dissolution and full immersion of solids, together with variable amounts of KOH pellets (≥85 %, Merck). Reactions were conducted at 220 °C for 4 h in a ventilated oven; residence time was measured from when the oven reached the setpoint (~25 min heating time). After cooling in a water bath, slurries were opened, pH measured, and products recovered. The temperature of 220 °C was chosen based on literature as optimal synthesis temperature (Marzban et al., 2024; Queffelec et al., 2025), while 4 h ensured the occurrence of secondary reactions and repolymerization. Each run was performed at least in triplicate for reproducibility. HTC conditions correspond to 0 % KOH.

2.2. Recovery of hydrothermal products

A-HS were divided into humic acid (HA), fulvic acid (FA), and humin (HU) following IHSS guidelines (Queffelec et al., 2025). HA is the solid phase recovered by acidic precipitation (pH 1.5, HCl) of the liquid phase; FA is the soluble phase after this step; HU is the insoluble solid obtained under HTH conditions (ending neutral/alkaline). Under acidic HTC, the solid product is referred to as hydrochar (HC). Slurries from hydrothermal runs were diluted with 7 mL deionized water (to minimize A-HS retention on solids), centrifuged at 9000 rpm for 10 min (Hettich Universal 320, Germany), and filtered through 0.45 µm membranes (Whatman). The solid fraction (HU or HC) was dried at 105 °C overnight and weighed. HA and FA were separated using a modified IHSS protocol: the liquid phase was acidified to pH 1.5 with 1 M HCl, centrifuged (9000 rpm, 10 min), and the precipitate (HA) collected, dialyzed in 3.5 kDa cellulose membranes until conductivity < 0.1 µS, freeze-dried (24 h, LYOTECH GT2-E), and weighed. The FA-containing supernatant was extracted on preconditioned Bond Elut PPL cartridges (Agilent, USA, 200 mg), washed with 1 M HCl, air-dried, and eluted with methanol; eluates were evaporated by rotary evaporation and stored. Solid yield (SY) was computed as the percentage ratio between the mass of the solid phase (either HC, HU, or HA after acidic precipitation) and initial biomass on a dry base.

2.3. Characterization of the solid and liquid phases

Thermogravimetric analyses (TGA) were performed on a TG 209 F1 Libra (Netsch, Germany), under high-purity N₂ (20 mL/min) at 10 °C/min from 25 to 900 °C. The extent of converted sample was calculated as $x(t) = (m_0 - m_t)/(m_0 - m_f)$, where m_0 is the dry initial mass, m_t the mass at time t , and m_f the residual mass at 900 °C. Derivative curves (DTG)

were obtained by plotting dx/dt vs temperature, and residual mass fraction (TGA_{res}) defined as m_t/m_0 . Ash content was determined by oxidation in a muffle at 950 °C. Elemental composition was determined with a vario-MICRO cube CHNOS analyzer (Elementar Analysensysteme GmbH, Langensfeld). C, H, and N were quantified with a thermal conductivity detector (TCD), while O was calculated by difference. Morphology was examined with a LEO 1550-Gemini Zeiss SEM at 3 kV after gold coating. FTIR spectra were collected on a Nicolet iS 5 spectrometer with an iD5 ATR unit (Thermo Fisher Scientific) over 4000–400 cm^{-1} , at 2 cm^{-1} resolution and 32 scans for both sample and background. XPS was performed on a Thermo Fisher K-Alpha system with monochromatic Al K α radiation (1486.6 eV). Powders were mounted on carbon tape; survey scans were acquired at 200 eV pass energy (1 eV step, $\sim 400 \mu m$ spot) and high-resolution C 1 s spectra (280–298 eV) at 50 eV pass energy (0.1 eV step). Binding energies were calibrated to the C–C peak at 284.8 eV, and spectra analyzed with XPSPeak. Solid-state ^{13}C NMR analyses were performed on a Bruker BioSpin SOLIDS400WB 400 MHz spectrometer with an HX probe under MAS at 13.5 kHz, using a CP sequence with 1000 μs contact time. The chemical shift scale was referenced to glycine (176.5 ppm). Shifts were grouped into regions: I (0–60 ppm), II (60–110 ppm), III (110–170 ppm), IV (160–200 ppm). Optical properties were evaluated by UV–Vis and fluorescence spectroscopy. Absorption spectra were acquired on a T70 + UV/VIS spectrophotometer (PG Instruments, UK) in the 200–800 nm range using a 1 cm quartz cuvette. Excitation–emission matrices (EEMs) were collected with an FP-8300 spectrofluorometer (JASCO, USA), with excitation wavelengths of 250–450 nm and emission wavelengths of 280–600 nm, also in a 1 cm quartz cuvette.

The liquid phase was analyzed by HPLC and UHPLC. Aromatics (phenols and furans) were quantified with an HPLC system (ICS-3000, UV detector, Dionex/Thermo Fisher), after 1:4 dilution in 10 % acetonitrile and filtration through PTFE syringe filters (0.2 μm , Neolab). Carboxylic acids (acetic, lactic, formic) and carbohydrates (glucose, fructose) were measured with a UHPLC (Ultimate 3000 Dionex, Eurokat H column, 300 mm $\times$ 8 mm, 10 μm , KNAUER) using 0.01 N H_2SO_4 as mobile phase. Relative peak areas were grouped into carboxylic acids, furans, and phenolic derivatives. 1H and HSQC NMR spectra were acquired on an Agilent 400 MHz spectrometer (101 MHz for ^{13}C). FA samples were dissolved in d_6 -DMSO. 1H spectra were recorded with 1 s delay and 128 scans, HSQC with 2 s delay and 32 scans. Spectra were processed with Mestrenova Software. Solvent signals (d_6 -DMSO 2.5 ppm, methanol 3.2 ppm, water 4.8 ppm) were excluded. Four regions were integrated: I (<2 ppm, alkyl), II (2–3 ppm, aliphatic), III (3–4.3 ppm, oxygenated aliphatic), IV (6–10 ppm, phenolic/furfural).

3. Results and discussion

3.1. Trends of final pH and yields

Fig. 1 shows the pH and SY of the residual solid (HU in HTH, HC in HTC) as a function of KOH concentration for different substrates. The final pH is always lower than the initial, indicating the formation of acidic species that react with alkali until eventual neutralization. When ending alkali or neutral (so in HTH), substrates undergo almost entire conversion, whereas conversion decreases progressively as the system shifts toward acidic HTC conditions.

Glucose and cellulose reach similar final pH values at the extremes: 9.7 under strongly alkaline and 2.7 under acidic conditions (Fig. 1a). Their SY–KOH trends show three segments with distinct slopes, reflecting the chemistries of alkaline vs acidic regimes. At high KOH, substrates are almost fully converted in the liquid phase, with negligible residual HU (SY of 1.1 % for glucose and 10 % for cellulose), suggesting the alkali hydrolysis is highly favored (Esposito and Antonietti, 2013). As KOH decreases, acidic species neutralize the alkali, shifting the reaction equilibrium. Neutral conditions (pH 6.5–7) occur at a KOH of 44.0 % for glucose and 38.5 % for cellulose, a difference attributed to the lower dissolution yield of cellulose compared to glucose, which results in fewer acidic species being neutralized. When reaching neutral conditions, the SY is 7.7 % for glucose, indicating that acidic pathways become more involved, while for cellulose, the number of unreacted materials increases. As KOH further decreases, SY drastically increases, from 7.5 to 40 % as the final pH drops from 7.5 to 4 in the case of glucose, in concomitance with the formation of a polyfuranic char. Further KOH reduction causes only minor SY changes, indicating that the main repolymerization reactions have already occurred. This phenomenon is consistent with their higher-order kinetics driven by collision and condensation, as polymerization reactions are often characterized by second-order kinetics, meaning their rate increases more than proportionally with the concentration of reagents (Jung et al., 2020). Cellulose exhibits a similar trend, with its SY increasing rapidly from 37.0 to 84.0 % as the pH drops from 8 to 5.5. Under acidic-HTC, the conversion is almost absent, with a SY of 95.6 %. In this case, acidic hydrolysis controls the conversion, though it is less effective than the alkaline process. Indeed, H^+ ions are less efficient than OH^- ions at disrupting the crystalline structure of cellulose, as they interact less strongly with the hydrogen bonds within cellulose chains. Achieving similar conversion yields under acidic would also require more severe temperatures (>250–280 °C) (Toor et al., 2011).

Lignin requires a higher alkali concentration to undergo a significant conversion, with a KOH higher than 20 % (pH > 9) to cause a conversion

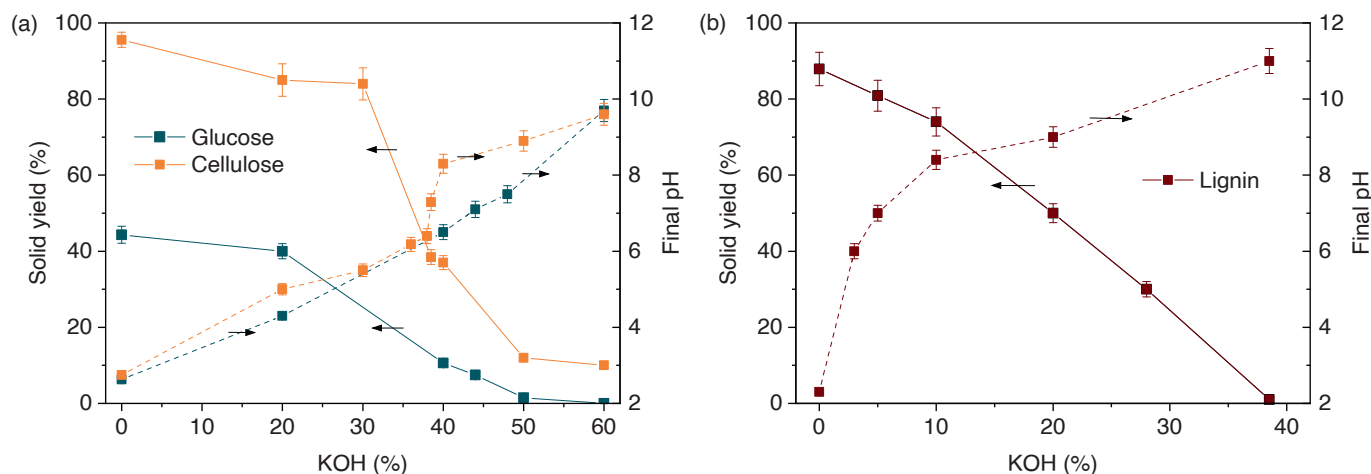


Fig. 1. Effect of KOH on solid yield (full lines) and final pH (dashed) for: a) glucose and cellulose; b) lignin. KOH is expressed on a mass% basis relative to the initial biomass. A KOH of 40, 30 and 5 % corresponds to initial pH of 14.0, 13.0, and 10, respectively.

of over 50 % (Fig. 1b). Its β -O-4 ether bonds and aromatic structure confer high stability, and high concentrations of OH^- ions are needed to cleave them and deprotonate phenolic hydroxyl groups, leading to the solubilization of the resulting fragments (Schutyser et al., 2018). The capability of alkali to breakdown lignin is known from the literature and is widely used in traditional pulp extraction (Figueiredo et al., 2018). Under a mild alkali or HTC regime, the conversion is mild, achieving a SY of 87.8 %. Higher temperatures (>300 – 320 °C) would be required to achieve some more significant conversion (Toor et al., 2011), and the amount of acids released is too low to promote self-catalytic hydrolysis, which is also reflected in the smaller amount of neutralizing KOH (5.0 %) compared to carbohydrates. SY-KOH trends can be divided into two regions based on the pH slope. Below KOH of 10 %, each increment in KOH leads to a pronounced pH increase, indicating rapid neutralization of OH^- by the acidic species released from lignin. Above 10 %, the buffering capacity of these acidic species likely becomes saturated. While additional lignin dissolves, it does so as larger fragments, which are less effective at neutralizing the added KOH, resulting in a slower pH increase and a flattening curve.

3.2. Aqueous phase composition

HPLC data (Fig. 2 and Fig. S1) show the evolution of aqueous-phase compounds, grouped as carboxylic acids, phenols, and furans. Their concentrations rise with alkalinity, consistent with the greater biomass conversion into the liquid phase – reaching up to 100.1 g/L for glucose, 75.3 g/L for cellulose, and 8.6 g/L for lignin under the most alkaline conditions (Table S1).

For both glucose and cellulose, carboxylic acids dominate the aqueous phase, representing > 96.6 % of quantified compounds. Although the absolute pathways differ between alkaline and acidic regimes, this highlights that organic acids remain the prevailing products in both cases. Under HTH, lactic acid is measured. It is a typical signature of alkaline digestion, known to form via *retro*-aldol cleavage of glucose, passing via intermediates like glyceraldehyde and pyruvaldehyde (Esposito and Antonietti, 2013). In the case of cellulose, the path is the same but anticipated by the hydrolysis of cellulose to glucose – reaction favored by alkaline conditions, where OH^- ions weaken cellulose's hydrogen bonds (reducing its crystallinity) and act as nucleophiles, attacking the glycosidic linkages and cleaving the β -1,4-glycosidic bonds (Argun and Onaran, 2016). Formic and acetic acid are also detected, which form from oxidation of lactic acid itself and as end products from oxidation of aldehydic intermediates of glucose. All these acids participate in neutralizing the alkali, lowering the pH. Although furans are not detected in the liquid phase under HTH, analyses on the residual solid (Section 3.5) reveal the presence of repolymerized

products, suggesting that furans form in traces and rapidly recondense. Interestingly, under HTH, phenols are detected, (around 2 % of the total) and are also present in the final HA structure (as discussed later). Phenols are likely secondary products forming via the aldol condensation of aldehydic intermediates: the alkali catalyzes *retro*-aldol cleavage reactions, forming carbonyl fragments that, through enolate intermediates, undergo aldol condensation and rearrangements to phenolic derivatives. In literature, they were observed after the hydrothermal treatment of carbohydrate-rich materials in the presence of Na_2CO_3 (Russell et al., 1983) or alkali salts (Kruse and Dinjus, 2005), which are a similar reaction environment with HTH. The prevalence of phenols over furans in HTH is favorable for humification, since they are typical motifs of natural HS and thus a signature of humic-like structures (Spaccini and Piccolo, 2009). They also correlate with redox buffering capacity, stimulating soil microbial activity and thereby improving nutrient cycling (Wells, 2019). In contrast, furans are dehydration products typical of acidic hydrothermal conditions, unstable at high pH and prone to recondensation into furanic-inert carbonaceous phases (i. e., the HC), which contribute little to the functional group characteristic of HS. Indeed, when transitioning to the HTC-acidic regime, 5-hydroxymethylfurfural (HMF) and furfurals form (Falco et al., 2011a), indicating that the *retro*-aldol pathway gives way to acidic hydrolysis and dehydration. In this acidic environment, acetic acid levels increase, primarily due to the oxidation of HMF, while formic acid and lactic acids (still detected) originate from HMF rehydration (Falco et al., 2011a).

In the case of lignin, significantly fewer compounds are identified compared to carbohydrates. The main reason is that high-molecular-weight lignin fragments remain undetectable by the HPLC method, resulting in an apparent predominance of carboxylic acids. Phenolic-derivatives are anyway detected in large quantities, deriving from the hydrolysis and cleavage of other bonds in lignin. Among them, guaiacol occupies the main fraction (up to 52.1 %), deriving from primary depolymerization of coniferyl alcohol. Guaiacol degradation leads to catechol, phenols and cresol, all detected. Instead, carboxylic acids derive from the oxidation of aromatic rings of lignin fragments (Toor et al., 2011), with acetic acid predominating and other acids likely deriving from carbohydrate impurities in the original lignin.

3.3. Humic acids from carbohydrates

HA is the solid fraction of A-HS recovered from acidic precipitation from the liquid phase, present only after HTH. HAs properties slightly vary among glucose and cellulose (Table 1). Their SY in the range of 1.1–3.2 %, similar to data in literature (Shao et al., 2023). SY increases with alkalinity mainly due to the higher dissolution degree of biomass under stronger alkaline conditions and a more intense extent of *retro*-

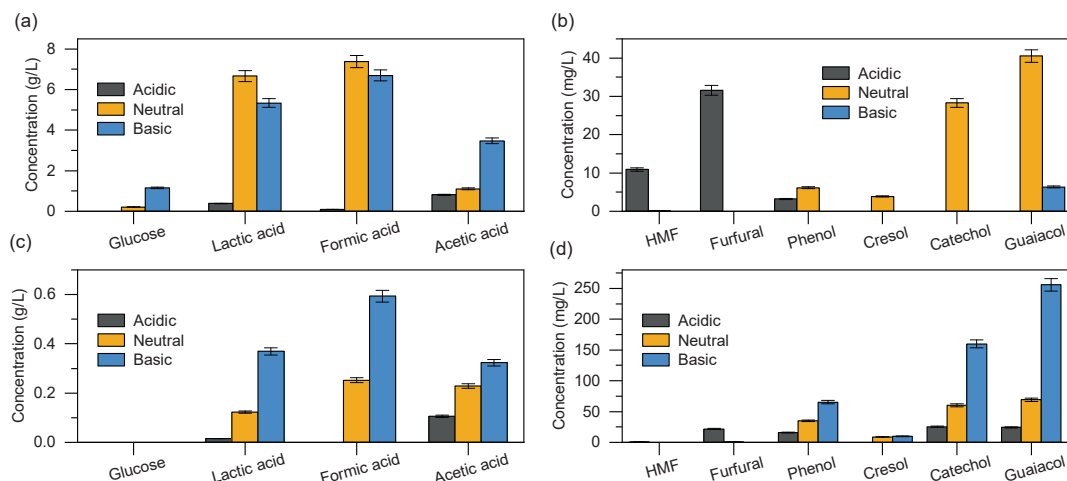


Fig. 2. HPLC-detected composition of the aqueous phase after hydrothermal treatment, as a function of final pH, derived: (a-b) from cellulose, (c-d) from lignin.

Table 1

Properties of the initial substrates and the corresponding humic acids obtained under different conditions. Data are organized by final pH regime, showing KOH dosage, SY, elemental composition (C, H, O), H/C and O/C atomic mass ratios, and TGA_{res}. Data represent mean values of three independent replicates. Standard deviations are reported. Acronyms: SY = solid yield, TGA_{res} = residual mass from TGA at 900 °C.

Substrate	Condition	KOH (%)	SY (%)	Ultimate analyses (wt.%)			H/C (–)	O/C (–)	TGA _{res} (%)
				C	H	O			
Glucose	Raw	–	–	39.2 ± 0.1	6.7 ± 0.1	54.1 ± 0.1	2.05 ± 0.03	1.04 ± 0.01	14.1 ± 0.1
	Neutral	44.0 ± 0.8	1.2 ± 0.1	64.6 ± 0.2	5.5 ± 0.1	30.0 ± 0.2	1.02 ± 0.02	0.35 ± 0.01	44.8 ± 0.2
	Alkaline	66.0 ± 1.0	3.1 ± 0.1	78.1 ± 0.2	5.9 ± 0.1	16.1 ± 0.2	0.91 ± 0.02	0.16 ± 0.01	47.1 ± 0.1
Cellulose	Raw	–	–	42.6 ± 0.2	6.5 ± 0.2	51.0 ± 0.3	1.83 ± 0.06	0.90 ± 0.07	12.4 ± 0.1
	Neutral	38.5 ± 0.5	1.1 ± 0.1	63.4 ± 0.3	5.8 ± 0.1	30.8 ± 0.3	1.10 ± 0.02	0.37 ± 0.01	50.9 ± 0.2
	Alkaline	60.0 ± 0.9	3.2 ± 0.1	76.2 ± 0.3	6.6 ± 0.1	17.3 ± 0.4	1.04 ± 0.02	0.17 ± 0.01	42.0 ± 0.1
Lignin	Raw	–	–	66.0 ± 0.4	5.7 ± 0.1	28.3 ± 0.4	1.04 ± 0.02	0.32 ± 0.01	42.6 ± 0.1
	Neutral	5.0 ± 0.2	10.9 ± 0.1	62.6 ± 0.3	5.2 ± 0.2	32.2 ± 0.4	1.00 ± 0.04	0.39 ± 0.01	49.0 ± 0.3
	Alkaline	38.5 ± 0.8	80.1 ± 0.5	79.6 ± 0.5	6.1 ± 0.1	14.2 ± 0.5	0.92 ± 0.02	0.13 ± 0.01	53.3 ± 0.2

aldol reactions, leading to a higher number of HAs precursors. SEM images (Fig. S2) reveal a typical morphology consisting of agglomerated, non-ordered structures, which is a typical fingerprint of that of natural HS (Spaccini and Piccolo, 2009). This morphology also agrees with their hydrophobic character: the progressive pH decrement deprotonates functional groups, minimizing repulsion forces and causing the molecular structure to collapse into a compact, coiled-like form (De Melo et al., 2016). The C content of HAs reaches 63.4–64.6 % in HA-neutral and 76.2–78.1 % in HA-basic, significantly increasing from the 39.3–42.3 % of the starting material. This increment is accompanied by an O decrement and a moderate increase in H, leading

to a reduction in the O/C and H/C ratios. The slight differences in O/C and H/C ratios between HA-neutral and HA-basic indicates that alkaline conditions promote O removal primarily via condensation rather than decarboxylation and aromatization. Consistent with elemental analysis, the DTG curves of HAs (Fig. S3) show improved thermal stability with the process, evidenced by a shift of the DTG peak to higher temperatures (from 315 to 425 °C for glucose and from 352 to 390 °C for cellulose) and a decrease in peak intensity. This observation suggests a significant structural reorganization of glucose/cellulose toward a more thermally stable material. In terms of pathway, O-rich intermediates—such as large aldehydic species—underwent aldol condensation into larger and

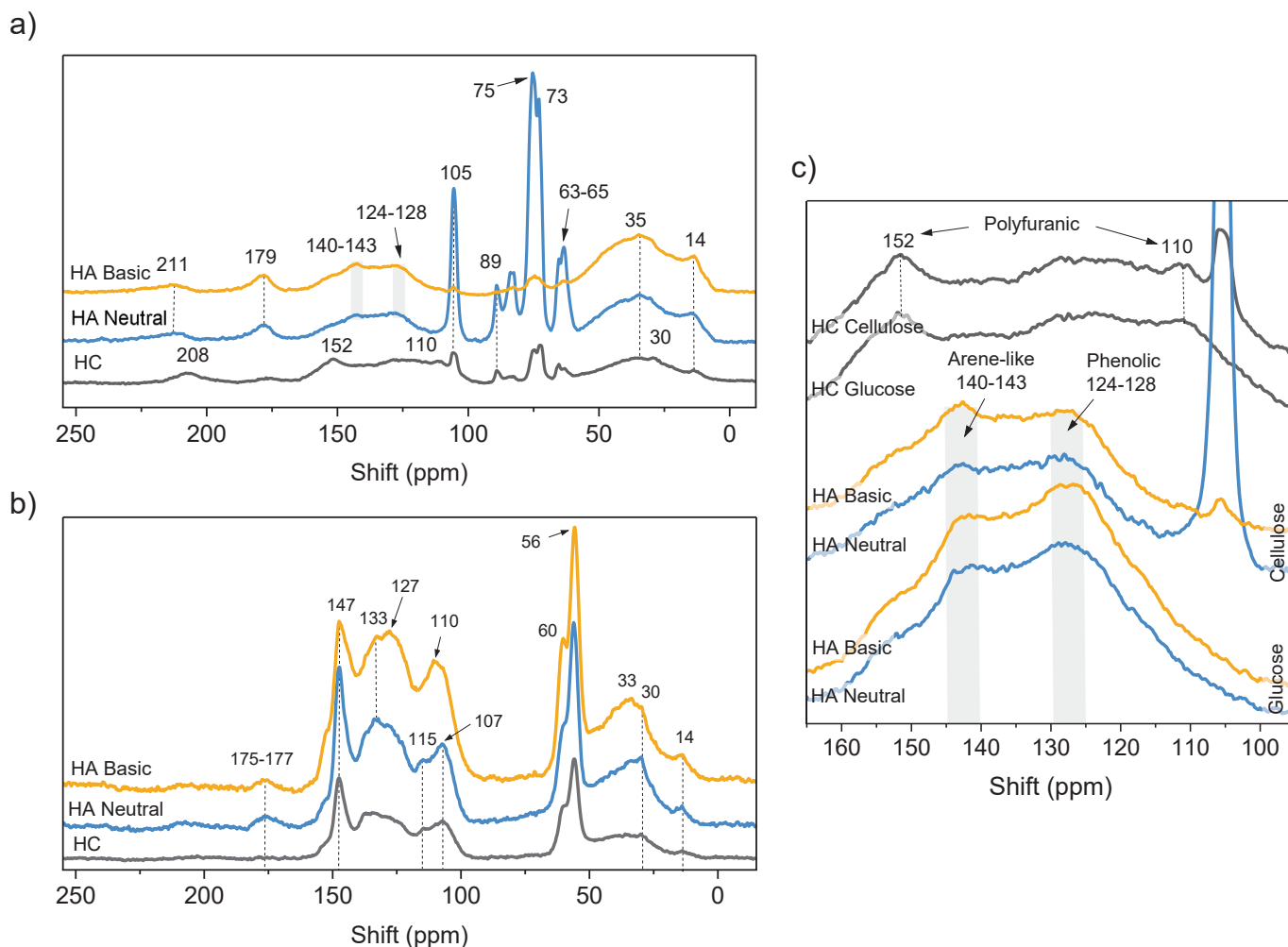


Fig. 3. Solid-state ^{13}C NMR spectra of HA and HC obtained under different final pH conditions: a) from cellulose; b) from lignin; c) zoom-in of the aromatic region for samples derived from glucose and cellulose.

more condensed carbon frameworks.

FTIR spectra (Fig. S4) show HAs condensation structure. Indeed, the band at 1615 cm^{-1} corresponds to aromatic C=C vibrations, while the bands at 1460 and 1425 cm^{-1} to C—H vibrations of methoxy groups. The band at 2900 cm^{-1} indicates aliphatic structures associated with CH_3 and CH_2 stretching (Wei et al., 2022). O-groups are evidenced by bands at $3000\text{--}3500\text{ cm}^{-1}$ of —OH groups, $1695\text{--}1700\text{ cm}^{-1}$ due to the stretching of —COOH and C=O, and $1030\text{--}1200\text{ cm}^{-1}$ to the stretching of C—O (Wei et al., 2022). O-groups confer the hydrophilic character of HAs, similar to natural HA. HA spectra are nearly identical across samples, mainly due to the surface nature of FTIR analyses and the inherent surface similarity across samples. Similarly, XPS spectra (Fig. S5) show that HAs have signals attributed to aliphatic/aromatic carbon (C—C/C=C at 284.6), ether/alcohol carbon (C—O at 285.9 eV), carbonyl groups (C=O at 287.3 eV) and carboxylic groups (O=C—O at 289.3 eV) (Marzban et al., 2024). There is a predominance of C—C/C=C groups, which occupy an area up to 78.0% (Table S2) for glucose, followed by C—O and C=O groups. The higher presence of O groups in HA-neutral agrees with the presence of cellulose remains.

Solid-state ^{13}C NMR spectra of HAs (Fig. 3 and Fig. S6) show the typical aliphatic signature in the $0\text{--}60\text{ ppm}$ region, occupying the main relative areas (up to 67.2%). The broad hump at $30\text{--}40\text{ ppm}$ is due to disordered CH_2 groups in long-chain alkyl structures (Falco et al., 2011a), likely deriving from aldol condensation products forming branched aliphatic—aromatic networks. The broadness of the signal reflects a cross-linked and heterogeneous structure characteristic of HS (Soares da Silva et al., 2020). The 14 ppm signal can be attributed to CH_3 groups at the ends of alkyl chains reflecting a rich alkylated composition. Samples from cellulose show unreacted cellulose in the $60\text{--}110\text{ ppm}$ region, with $63\text{--}65\text{ ppm}$ C_6 monomeric units, $84\text{--}89\text{ ppm}$ C_4 monomeric units, $73\text{--}75\text{ ppm}$ unreacted glucose (Falco et al., 2011a). Neutral-HA supports the previous observation that residual cellulose is retained within HAs, exhibiting more prominent peaks, which nearly disappear in HA-basic, leaving only two peaks at 75 and 105 ppm . This difference is further highlighted by the relative areas (Table S3), with Region II accounting for 51.1% in neutral conditions vs 2.4% under alkaline conditions. Zooming in the aromatic region ($110\text{--}170\text{ ppm}$, Fig. 3c), HAs display a characteristic polyaromatic signature. Indeed, the $140\text{--}143\text{ ppm}$ band indicates a developed polyaromatic arenic-like scaffold, while the $124\text{--}128\text{ ppm}$ band phenolic carbons close to quaternary and protonated sp^2 carbons (Falco et al., 2011a). This observation agrees with the HTH liquid-phase composition containing phenols but no furans. It is reasonable that the A-HS formation mechanism may involve aldehydic intermediates that undergo aldol condensation, cyclization, and dehydration to yield phenolic compounds or aromatic aldehydes, which are incorporated into the HA structure. Indeed, aldehydic intermediates are known precursors in alkali hydrothermal reactions (Esposito and Antonietti, 2013), as well as in their subsequent aldol condensation steps, which form C—C bonds and phenolic/aromatic domains (Kruse and Dinjus, 2005). Although we do not directly measure aldehydes, the evidence of the liquid composition, solid-state NMR showing phenolic domains, and literature support this pathway as a probable mechanism for aromatic ring formation during HTH. The signal at 179 ppm can be attributed to COOH groups or quinone carbonyl groups, which are markers of the oxidation and aromaticity of HAs (Kida et al., 2019). The peak at 211 ppm in HAs indicates the presence of highly oxidized carbonyl groups, likely associated with functional groups such as carboxyl or quinone. This slight shift compared to HC, which commonly shows the carbonyl peak at 208 ppm (Falco et al., 2011b), suggests that HAs have undergone more extensive oxidation, leading to a higher degree of deshielding of the carbonyl carbon, contributing to a more O-rich structure. Generally, the presence of these aromatic cores is key for carbon sequestration, promoting persistence and the formation of stable humic-clay complexes that protect organic matter in soils (Cao and Schmidt-Rohr, 2018).

HAs are fluorescent, as shown in absorption and EEM spectra

(Fig. S7). They exhibit an exponential pattern typical of natural HS, characterized by increasing absorption at shorter wavelengths and an extended tail in the visible region, which indicates a high degree of conjugation and smooth electron transfer. EEM spectra reveal similar trends for glucose and cellulose HS under neutral and alkaline conditions, identifying two fluorescence regions: A ($\lambda_{\text{ex}} 250\text{--}300\text{ nm}$, $\lambda_{\text{em}} 340\text{--}550\text{ nm}$) and B ($\lambda_{\text{ex}} 320\text{--}400\text{ nm}$, $\lambda_{\text{em}} 400\text{--}550\text{ nm}$), corresponding to low- and high-molecular-weight humic fractions, suggesting a significant presence of phenol-like compounds (Soares da Silva et al., 2020). HAs from glucose do not show significant differences among neutral and basic conditions. In the case of cellulose, the basic-HA shows a shift of emission peak A to shorter wavelengths ($\sim 320\text{ nm}$), possibly indicating increased formation of low-molecular-weight fluorescent molecules, such as phenols.

3.4. Humic acids from lignin

Alkali strongly enhances the conversion efficiency of lignin to HA, with its SY passing from a SY of $10.9\text{--}80.1\%$ when KOH is increased from 5 to 30% KOH (Table 1). HA-neutral slightly differs from the original lignin. It shows a slight decrease in C and its DTG peak remains at 364°C as lignin with reduced intensity, indicating a partially oxidized structure with a slightly improved thermal stability. Instead, HA-basic shows a significant C increase of up to 79.6% . The similarity of its H/C ratio to lignin, the O/C drop to 0.1 , and the slight variation in the DTG peak indicate that harsher alkalinity enhances the removal of O functionalities and promotes some aromatic condensation while preserving the initial lignin structure. Similarly, FTIR spectra (Fig. S4) show that both HAs retain the main lignin structural moieties, clearly maintaining the aromatic C—C and C=C (~ 1600 and 1512 cm^{-1}), C—H vibrations (~ 1460 and 1425 cm^{-1}), and the symmetric and asymmetric C—H stretching of — CH_2 — and methyl — CH_3 groups (2940 and 2840 cm^{-1}) (Figueiredo et al., 2018). The main difference from lignin is the more intense C=O band ($\sim 1690\text{ cm}^{-1}$), reflecting the enrichment of HAs in oxygen-containing functionalities. Oxidation is also evidenced by XPS spectra (Fig. S5), which reveal that lignin-derived HAs exhibit prominent signals for C—O and C=O groups, dominating over C=C signals.

Solid-state ^{13}C NMR spectra (Fig. 3b) confirm that HAs retain the core structure of lignin. Indeed, the sharp signals at 56 and 147 ppm are due to methoxy carbons and C—O of aromatic rings, respectively, deriving from lignin (Crestini et al., 2017). HAs also retain the $175\text{--}177\text{ ppm}$ band, corresponding to C=O carbonyl groups of lignin, particularly more present in HA-neutral. HA-neutral preserves the 115 ppm signal, associated with the C—H of the C_5 aromatic ring in lignin, as well as the 14 ppm signal, likely originating from CH_3 aliphatic chains (Falco et al., 2011a). Then, the absence of the 74 ppm signal, associated with the α position in $\beta\text{--O-4}$ bonds of lignin, suggests that these ether linkages are cleaved during HTH and do not reform in the resulting HAs. Indeed, HAs appear as condensed lignin fragments, likely formed through C—C bonding at an extent of condensation increasing under alkaline conditions. This idea is supported by the emergence of a 127 ppm signal, attributed to protonated aromatic carbons (C_2 , C_5 , C_6) in condensed structures, and a 133 ppm signal, associated with C_1 aromatic carbons and oxygenated aromatic carbons, such as $\text{C}_4\text{—O}$, phenols, and quinones (Crestini et al., 2017). Notably, HA-basic exhibits a more prominent 127 ppm signal, whereas HA-neutral shows a more pronounced 133 ppm signal, aligning with the argument that milder HTH conditions limit condensation, preserving oxygen functionalities at C_1 and $\text{C}_4\text{—O}$ sites. In contrast, alkaline conditions promote the removal of oxygen functionalities, leading to extensive ether bond cleavage and oxidation loss, which enhances aromatic C—C condensation and drives the structure toward highly condensed polyaromatic domains (Falco et al., 2011a). Additionally, the 147 ppm peak in HA-basic is less sharp and intense than in HA-neutral, suggesting that stronger deoxygenation and aromatic condensation have reduced phenolic and oxygenated carbon content. A similar trend is observed in the aliphatic region, where the

aliphatic peak shifts from 33 ppm in HA-neutral to 30 ppm in HA-basic, indicating partial cleavage, oxidation, and reorganization of lignin's aliphatic side chains into shorter or functionalized fragments. Moreover, HA-basic exhibits a slightly less pronounced peak at 60 ppm, suggesting that while oxygen loss from aromatic rings occurred, some oxygenated aliphatic structures either persisted or were reformed. Similar to carbohydrate-derived HAs, the band at 175–177 ppm is attributed to COOH groups and serves as a signature for COOH groups or quinone carbonyl groups. Regarding soil-related applications, the retention of lignin-derived methoxy and oxygenated aromatic groups is relevant, since these moieties contribute to the redox activity of A-HS and support interactions with metals and nutrients. In particular, HA-neutral can offer higher reactivity through preserved oxygenated sites, whereas HA-basic favors long-term stabilization through aromatic condensation (Wei et al., 2022).

Finally, HAs exhibit a single fluorescence region in the EEM spectrum (Fig. S7, λ_{ex} 250–350 nm, λ_{em} 340–500 nm), indicating the presence of one type of fluorophore in the composition, which is likely related to residual aromatic rings from lignin (Crestini et al., 2017). The spectra are similar to those of carbohydrate-derived HAs but with a more pronounced shoulder at 280 nm, which derives from lignin itself and highlights the higher number of isolated aromatic fragments within the HA, persisting after HTH. Moreover, there are no significant differences in absorption between neutral and basic HAs, further supporting the idea that the fluorescence is primarily attributed to the same molecules, predominantly lignin-derived aromatic groups.

3.5. Properties of humins and hydrochars

In the case of glucose, the SY of HU is minimal (up to 7.7 %) and appears as an intermediate between glucose and repolymerized products (SEM in Fig. S2). Its C content of 67.7 % (vs glucose 39.3 %) and flattening and shift of the DTG curve towards higher temperatures (Fig. S3) confirm the structural reorganization of glucose. Cellulose HUs retain the original morphology, composition (Table 2), and FTIR spectra (Fig. S4) show no significant carbonization. The slight anticipation of the DTG peak to 330 °C is likely due to the cleavage of glycosidic bonds under alkaline conditions, reducing the residual cellulose's thermal stability. Additionally, XPS spectra (Fig. S5) reveal a lower relative abundance of C—C/C=C bonds than HA, with a more significant presence of O-containing bonds, again attributed to the retention of cellulose-like characteristics. When operating in the HTC regime, carbonization becomes prominent for both glucose and cellulose. Indeed, HCs show an improved thermal stability compared to HU, as exhibited by the C content of 67.7–69.3 % and the DTG peak shift to 425 °C with reduced intensity. Glucose HC is well known to be formed by repolymerized HMF undergoing carbonization (Falco et al., 2011a). Similarly, cellulose HC consists of a charred phase derived from the solid–solid conversion of cellulose, along with repolymerized products from the liquid phase. Compared to the corresponding HAs, HCs display

a similar DTG peak position (~425 °C) but slightly lower peak intensity (Fig. S3) and a higher TGA_{res} (Tables 1 and 2). This observation reflects the fact that HC originates from repolymerized furanic-type intermediates formed under HTC, which also undergo progressive carbonization during the hydrothermal step. As a result, HC yields a higher final residue in TGA (e.g., 66.5 % for glucose HC vs. 44.8–47.1 % for the corresponding HAs, Table 2). In contrast, HAs, although rich in phenolic and arenic structures, contain more oxygenated functional groups (carboxyl, carbonyl, phenolic OH), which make HAs chemically reactive (and environmentally attractive), but they also decarboxylate and volatilize during heating, slightly lowering thermal stability. This interpretation is consistent with the slightly different H/C and O/C ratios between HA and HC, where HA retains higher oxygen content and functional diversity despite a lower thermal residue.

XPS spectra show predominant relative areas occupied by C—C/C=C bonds. FTIR spectra of HCs closely resemble those of HAs, though with less intense C=O bands and weaker C—H/C—H₂ bands, which is likely attributable to the tendency of HC to adsorb carboxylic acids, thereby altering their surface rather than a structural similarity to HA. Further supporting this theory, solid-state ¹³C NMR spectra of HCs (Fig. 3a) shows how HCs keep their typical polyfuranic signature, with two resonance peaks at 150 ppm and 110 ppm, as observed in the literature (Falco et al., 2011b), clearly distinguishing from HAs. Moreover, HCs display sharper and more defined signals at 37 and 30 ppm than HAs. The 37 ppm signal corresponds to CH₂ groups in shorter aliphatic chains, likely attached to the polyfuranic framework, while the 30 ppm signal is attributed to CH₃ groups or shorter CH₂ units in alkyl fragments, possibly from terminal methyl groups or small, isolated aliphatic chains (Falco et al., 2011a).

In the case of lignin, the residual HU resembles lignin in terms of composition (Table 1), thermal stability (Fig. 2c), and morphology (Fig. S2), indicating its mild degradation during the conversion. FTIR spectra of HU resemble lignin apart from shifted peaks at 2980 and 2900 cm⁻¹, suggesting altered aliphatic structures likely arising from a mild depolymerization and reorganization of lignin's side chains. Similarly, HU retain a predominantly aliphatic/aromatic signature, as shown by the strong C—C/C—H signals, highlighting their closer resemblance to the original lignin framework. Shifting towards acidic conditions (entering HTC), lignin undergoes minimal mass loss (~13 %), with HC showing a composition and DTG curve similar to raw lignin. More severe conditions would be needed for some substantial re-organization (Rodríguez et al., 2014). The absence of carbonization is well shown by the ¹³C NMR spectra (Fig. 3b). The 147 ppm signal is retained, reflecting the persistence of lignin C—O in aromatic rings. Compared to the HA, the 175–177 ppm band indicates a lower presence of C=O carbonyl groups due to a lower oxidation of HC. The undefined signals at 133 and 127 ppm indicate the lack of significant aromatic condensation. The lower intensity at 56 ppm suggests a reduction in methoxy groups, while the less pronounced peaks at 30 and 33 ppm differences in aliphatic side chains. These findings reinforce that lignin undergoes

Table 2

Properties of HC and HU obtained from glucose, cellulose, and lignin under different final pH regimes. Data include SY, elemental composition (C, H, O), atomic ratios (H/C and O/C), and residual mass from TGA (TGA_{res}). Values represent mean ± standard deviation (n = 3). Acronyms: HC = hydrochar (acidic HTC regime); HU = humin (neutral/alkaline HTH regime); SY = solid yield; TGA_{res} = residual mass from TGA at 900 °C.

Substrate	Condition	SY (%)	Ultimate analyses (wt.%)			H/C (–)	O/C (–)	TGA _{res} (%)
			C	H	O			
Glucose	HC Acidic	44.3 ± 1.0	67.9 ± 0.1	5.6 ± 0.1	26.6 ± 0.1	0.99 ± 0.02	0.39 ± 0.01	66.5 ± 0.1
	HU Neutral	7.7 ± 0.3	67.7 ± 0.2	5.6 ± 0.1	26.8 ± 0.2	0.99 ± 0.02	0.39 ± 0.01	60.9 ± 0.1
	HU Basic	1.5 ± 0.1	—	—	—	—	—	—
Cellulose	HC Acidic	50.2 ± 0.9	69.3 ± 0.3	4.6 ± 0.1	26.2 ± 0.3	0.80 ± 0.02	0.30 ± 0.01	49.5 ± 0.1
	HU Neutral	38.5 ± 0.2	46.2 ± 0.2	6.5 ± 0.2	47.4 ± 0.3	1.70 ± 0.06	0.80 ± 0.01	19.3 ± 0.1
	HU Basic	10.5 ± 0.2	44.9 ± 0.4	6.5 ± 0.1	48.7 ± 0.4	1.73 ± 0.04	0.81 ± 0.02	20.5 ± 0.1
Lignin	HC Acidic	87.9 ± 0.8	68.9 ± 0.4	5.6 ± 0.2	25.7 ± 0.5	1.00 ± 0.02	0.37 ± 0.01	46.2 ± 0.1
	HU Neutral	80.9 ± 1.0	68.8 ± 0.5	5.3 ± 0.1	25.9 ± 0.5	0.92 ± 0.02	0.35 ± 0.01	56.0 ± 0.1
	HU Basic	1.1 ± 0.1	—	—	—	—	—	—

minimal reorganization under acid-HTC and retains its original structure, unlike HA, which exhibits O loss and condensation under alkaline conditions.

3.6. Properties of fulvic acids

FA is the fraction of A-HS remaining soluble after acidification. ^1H NMR and HSQC spectra of all FAs (Fig. S8) are not well-resolved, which is a key sign of their inherent complexity as mixtures of low- and high-molecular-weight species (Spaccini and Piccolo, 2009).

Regarding FAs from glucose and cellulose, ^1H NMR spectra show that they mainly consist of alkyl (Region I) and aliphatic protons (Region II), which together occupy more than 80 % of the relative areas. The observation indicates that FA formation involves significant depolymerization and oxidation of carbohydrate intermediates, favoring the retention of low-molecular-weight, non-aromatic species (Queffelec et al., 2025). No significant differences are observed among the samples. HSQC spectra show that they contain methyl and methylene groups in saturated hydrocarbon chains (^1H shifts at ~ 0.9 – 1.5 ppm and ^{13}C signals at 10–30 ppm), which likely derive from fragmented carbohydrate-derived precursors, as well $-\text{CH}_2$ signals (20–40 ppm in ^{13}C and ^1H shifts at ~ 1.2 – 2.5 ppm) that are likely branched or linear aliphatic chains, possibly linked to oxygenated functionalities (Halleraker and Barth, 2020). Samples show the presence of oxygenated aliphatic protons in Region III. While for glucose they likely derive from partially oxidized carbohydrate fragments, like hydroxylated aldehydes or carboxylated intermediates, for cellulose they more probably consist of products from incomplete hydrolysis. Indeed, for cellulose-FA, they occupy a larger fraction for FA-neutral (up to 14.2 %), consistent with the retention of partially degraded cellulosic remains. Then, all samples present phenolic derivatives: HSQC spectra show aromatic protons correlated at aromatic carbons (6.5 ppm for $^1\text{H} \leftrightarrow 115$ ppm for ^{13}C) associated to oxygenated carbons typical of phenolic groups in aromatic rings, and 7.0–7.7 ppm for $^1\text{H} \leftrightarrow 120$ – 130 ppm for ^{13}C correlation typical of hydrogenated carbons in conjugated aromatic systems (Halleraker and Barth, 2020). These patterns exclude the presence of furfural derivatives, as furfural-type signals usually appear in the ^{13}C range at 140–160 ppm, which is absent here (Queffelec et al., 2025).

^1H NMR spectra of lignin show the predominance of Regions III and IV (> 80.5 %). HSQC spectra indicate benzylic hydroxylation at the $\text{C}\alpha/\text{C}\gamma$ positions of partially degraded β -O-4 linkages (3.0–4.0 ppm $^1\text{H} \leftrightarrow 30$ – 40 ppm ^{13}C), suggesting partial depolymerization with oxygenated side chains that hinder complete aromatic condensation (Halleraker and Barth, 2020). The correlation at 3.7–3.9 ppm $^1\text{H} \leftrightarrow 50$ – 60 ppm ^{13}C is due to methoxy groups attributable to syringyl and guaiacyl units. The region at 3.4–3.8 ppm $^1\text{H} \leftrightarrow 60$ – 70 ppm ^{13}C can be attributed to ether linkages and secondary alcohols, likely deriving from residual lignin ether structures or oxidized aliphatics (Halleraker and Barth, 2020). These features demonstrate that lignin-derived FAs maintain oxygenated aliphatic functionalities and methoxy groups, indicating that depolymerization occurs primarily through partial oxidation rather than complete aromatic condensation, in agreement with what observed for HAs. Region IV consists of phenolic groups within aromatic rings (as shown in HSQC 7.0–7.7 ppm $^1\text{H} \leftrightarrow 120$ – 130 ppm ^{13}C), which relative presence increases with alkalinity, achieving 43.4 %. This increase reflects enhanced oxidative cleavage of lignin's ether and β -O-4 linkages, releasing aromatic phenolic units that undergo further condensation and polymerization. Signals in regions I and II are also present, albeit to a lesser extent, suggesting that methyl and methylene groups from lignin's side chains and partially oxidized lignin fragments. In general, carbohydrate-derived non-aromatic species and phenolic compounds are helpful for soils functioning, since they are a nutrient source for soil microorganisms and support microbial activity (Mathew et al., 2021). Their solubility ensures nutrient availability and mobility, and the abundant oxygenated aliphatic protons favors nutrient binding and enhances the soil's cation exchange capacity, promoting the chelation of

metal ions and their bioavailability to plants (Spaccini and Piccolo, 2009).

3.7. Summarized reaction pathways and general considerations

Fig. 4a shows a summarized reaction pathway for glucose and cellulose according to the experimental data. Under HTH-alkaline conditions, they undergo hydrolysis and *retro*-aldol reactions, which become more intense under strong alkali. Glucose is cleaved into aldehydic intermediates and C_3 to C_1 compounds (e.g., lactic, acetic, and formic acids) via *retro*-aldol reactions, which eventually neutralize the alkali. Cellulose undergoes partial alkaline hydrolysis to yield glucose, while unconverted cellulosic fragments persist in solution due to their inherent recalcitrance. Phenols form as well, probably as secondary products via the aldol condensation of aldehydic intermediates: the alkali catalyzes *retro*-aldol cleavage reactions, forming carbonyl fragments that, through enolate intermediates, undergo aldol condensation and rearrangements to phenolic derivatives. Then, aldehydic intermediates undergo aldol condensation and cyclization, forming larger polyaromatic domains, yielding to A-HS. HAs (formation scheme in Fig. 4b) appear as polycondensed products rich in oxygenated functional groups, maintaining the aliphatic signature with polyaromatic structures (arenic-like and phenolic). Higher alkalinity favors condensed and carbon-rich HAs. FAs appear as a complex mixture of low- and high-molecular-weight compounds rich in alkyl and aliphatic protons with oxygen functionalities. HU is an intermediate repolymerized product for glucose, while for cellulose resembles chemically digested cellulose. When the reaction environment shifts to the acidic-HTC regime, the extent of conversion significantly decreases. Glucose transitions towards dehydration, yielding HMF and its back polymerization into a char-like phase (the HC) consisting of a polyfuranic framework. Cellulose hydrolysis is significantly limited and undergoes bulk reactions that preserve its original structure, resulting in an HC consisting of cellulose remnants.

In the case of lignin, during alkaline-HTH it undergoes significant defragmentation and/or depolymerization, probably driven by β -O-4 ether bond cleavage favored under alkaline conditions. Then, fragments undergo oxidative condensation, resulting in polycondensed and oxygen-rich A-HS (Fig. 4b). Both fragmentation and condensation are significantly enhanced by alkalinity. HAs appear rich in polycondensed structures and oxygenated functional groups yet retain most of the original lignin framework. Similarly, lignin-derived FAs retain oxygenated aliphatic functionalities and methoxy groups, maintaining the lignin core structure. Their solubility is favored by limited cross-linking into polyaromatic structures, as ether linkages and secondary alcohols persist, preventing extensive condensation. The lignin-derived HU fraction retains a chemical composition and thermal stability similar to lignin, with minimal oxidation and fragmentation. Under HTH-acidic conditions, lignin undergoes minimal conversion due to its complex polyaromatic structure. Indeed, acidic conditions do not effectively break ether and carbon-carbon bonds, and the resulting HC consists primarily of unreacted lignin, with only low yields of small aromatic fragments.

Beyond the mechanistic insights, it is worth noting that HTH offers several advantages over typical biotic processes like composting or anaerobic digestion. Indeed, HTH achieves humification within a few hours rather than weeks or months, with higher carbon efficiency and reduced GHG emissions, as the process does not lead to any gas production (CO_2 dissolves into the liquid phase under alkaline conditions). In potential scale-up scenarios, the thermal energy required to heat the slurry could be supplied using renewable (e.g., solar) or partially recovered by recirculating a fraction of the final slurry. While biotic pathways are advantageous for low technological complexity and costs, HTH offers an alternative for higher performance in terms of process control and carbon retention. However, the integration of abiotic and biotic humification pathways should not be excluded, particularly for particular biomass wastes where a biotic step could assist in

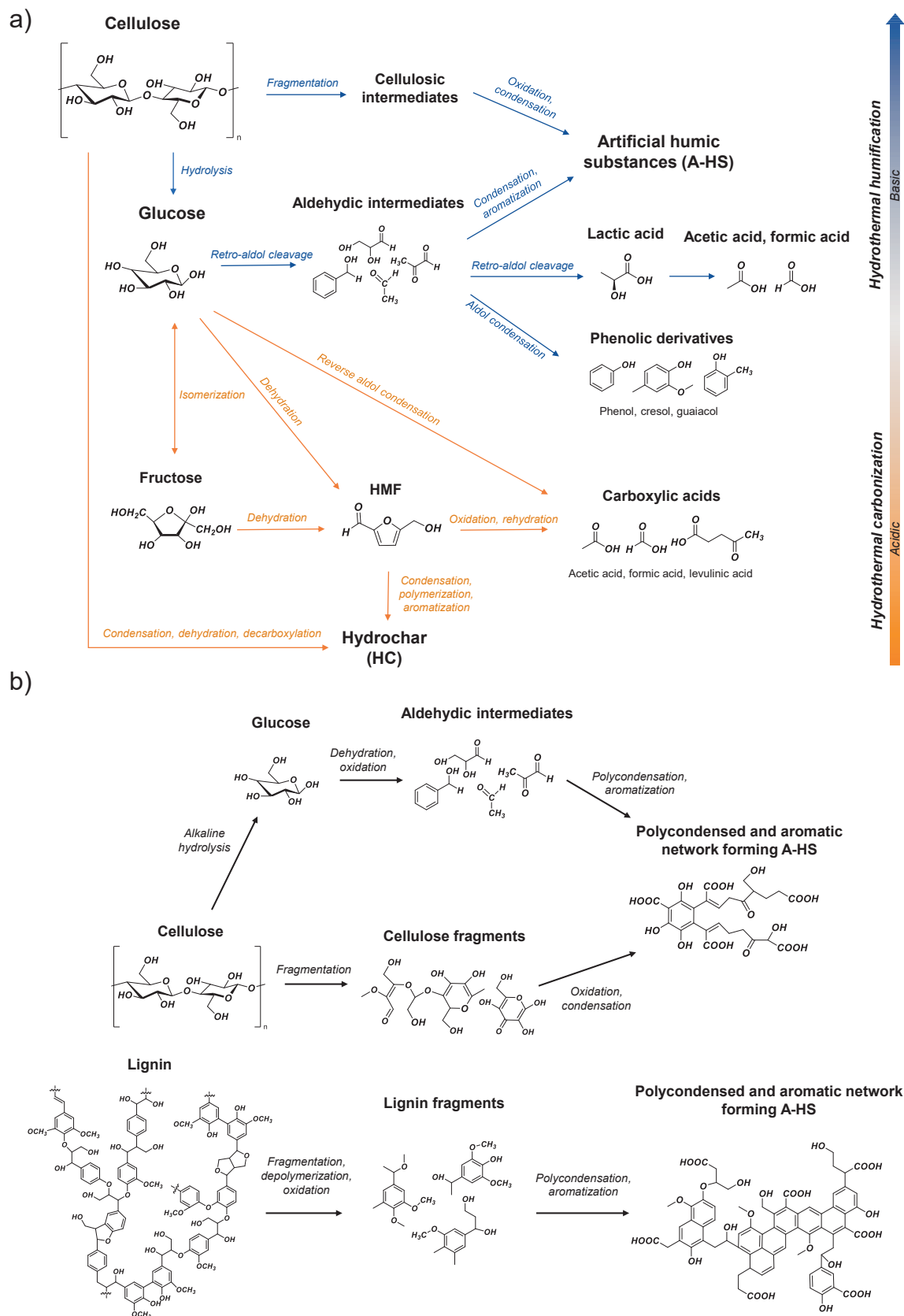


Fig. 4. a) hydrothermal conversion of carbohydrates, illustrating the transition from HTC (in orange) to HTH (in blue), passing from predominantly acidic or basic conditions; b) formation mechanisms of a-hs from hth of glucose, cellulose, and lignin.

pretreatment or stabilization. Moreover, while this study analyzed HA, FA, and HU separately for mechanistic clarity, in practice, these fractions can also be used in combination, especially when addressing heterogeneous wastes. This is a key difference from HTC, where the liquid phase is generally treated separately and only the HC is recovered, limiting the overall efficiency. Finally, it is worth noticing that the adoption of glucose, cellulose, and lignin as fundamental building blocks of lignocellulosic biomass provides a controlled framework to simplify the complexity of the reaction pathways governing the conversion of real feedstocks. While far from representing biomass heterogeneity, they are still a valuable starting point for deducing more complex mechanisms, and future kinetic studies could further deepen their reaction rates and pathways. Hemicellulose was excluded from this study due to its structural heterogeneity and variable solubility. Unlike lignin, whose heterogeneity mainly arises from differences in monolignol composition and linkages, hemicellulose composition varies strongly among biomass types (e.g., xylose-rich xylan in hardwoods, mannans in softwoods, arabinoxylans in cereals). Moreover, its general reactivity is reasonably represented by glucose (monomeric sugars) and cellulose (polysaccharidic backbone), but future work could address it for comprehensive mechanisms. In general, the presented results are relevant for the synthesis of A-HS, as they highlight how sugars, polysaccharides, and aromatics contribute complementary functionalities. The approach can be extended to biomass wastes with diverse properties, where combining substrates of different origins may enable the tailoring of A-HS targeting different properties and applications.

4. Conclusions

HTH is a novel route for converting biomass into A-HS under alkaline HTH, differing from classic acidic HTC that yields HC. In this study, model biomass compounds (glucose, cellulose, and lignin) are treated at 220 °C for 4 h under varying KOH concentrations to explore the transition from HTC to HTH. Under HTH-alkali, carbohydrates undergo *retro*-aldol cleavage into acids and aldehydic intermediates that condense into A-HS with an arenic-phenolic motifs. In contrast, shifting towards HTC-acidic, a polyfuranic HC forms. Lignin undergoes near-complete conversion, followed by condensation into A-HS, retaining its aromatic structure. In HTC, lignin conversion drops significantly. Overall, the resulting A-HS are rich in oxygen functionalities, show aromatic domains and fluorescence, properties that make them similar to natural HS, supporting applications for soil health, carbon sequestration, and biomass waste valorization. Results can serve as a starting point for investigating complex biomasses and tailoring A-HS for specific applications.

CRedit authorship contribution statement

Giulia Ischia: Writing – original draft, Validation, Software, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Nader Marzban:** Writing – review & editing, Validation, Methodology. **Johannes Schmidt:** Writing – review & editing, Methodology. **Alexander Volikov:** Writing – review & editing, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2025.133306>.

Data availability

Data will be made available on request.

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